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Author for correspondence:

Ubeiden C. Samboni

e-mail: ubeiden-samboni@unc.edu.ar

Alberto Sánchez López

e-mail: alberto.sanchez@unc.edu.ar

Theoretical insights into the electronic structure and NBO features driving nonlinear optical enhancement in the donor- π -acceptor system of OPPh₃py

Ubeiden C. Samboni,^{1,2} Luis Felipe Berrio Velasco,¹ Fernando Cuenú Cabezas,¹ Alejandro García Ríos,¹ Christian D. Lorenz³ and Alberto Sánchez López^{1,4}

¹Facultad de Ciencias Básicas y Tecnologías, Universidad del Quindío, Armenia - Quindío, Colombia

²Centro de Investigaciones en Química Biológica de Córdoba (CIQUIBIC), CONICET, Departamento de Química Biológica Ranwel Caputto, Facultad de Ciencias Químicas, Universidad Nacional de Córdoba, Ciudad Universitaria, Córdoba X5000HUA, Argentina

³Department of Engineering, King's College London, London, WC2R 2LS, United Kingdom

⁴Instituto de Investigaciones en Físico-Química de Córdoba (INFIQC), Departamento de Química Teórica y Computacional (DQTC), Facultad de Ciencias Químicas, Universidad Nacional de Córdoba, Ciudad Universitaria, Córdoba X5000HUA, Argentina

Organic materials capable of intramolecular charge transfer are of considerable interest for photonic, biophysics, medicine applications because of their strong nonlinear optical (NLO) responses and favourable electro-optic properties. In this work, diphenyl-[2-(2-pyridylaminomethyl)phenyl]phosphine oxide (OPPh₃py) was investigated as a potential NLO material using experimental characterisation and density functional theory (DFT) calculations. Structural and electronic properties were examined through molecular electrostatic potential analysis, frontier molecular orbitals, natural bond orbital (NBO) calculations, and intramolecular hydrogen-bonding interactions. Static and

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frequency-dependent NLO properties were evaluated at $\lambda = \infty$, 1906, 1341, 1064, 625, and 532 nm. The molecule exhibits pronounced HOMO–LUMO separation, strong donor–acceptor interactions, and a stabilising N38–H39 $\cdots$ O2 intramolecular hydrogen bond that promotes efficient charge transfer. NBO analysis further confirmed extensive charge delocalisation throughout the molecular framework. As a result, OPPh₃py displays a large dipole moment, high polarisability, and enhanced first- and second-order hyperpolarisabilities compared with the benchmark *p*-nitroaniline. These results demonstrate that OPPh₃py combines electronic stability with strong NLO activity, highlighting its potential for future advanced optical and electronic applications.

1. Introduction

Nonlinear optical (NLO) materials have attracted attention due to their broad applicability across multiple disciplines. Their applications include materials science [1–4], photonic and optoelectronic devices [5–7], biological sciences [8–10], biophysics [11,12] and medicine [13,14]. NLO materials include hybrid, polymer, organometallic, organic and inorganic compounds. Organic NLO materials are particularly attractive because of their structural versatility, low dielectric constants, mechanical flexibility, facile synthesis, broad availability, and high NLO efficiencies [15,16]. Inorganic NLO materials primarily derive their nonlinear response from noncentrosymmetric crystal structures and distorted coordination polyhedra, whereas organic NLO materials are dominated by π -electron delocalisation and intramolecular charge transfer (ICT) [17]. The structural flexibility of organic chromophores enables precise molecular engineering, allowing fine-tuning of electronic and optical properties to meet specific application requirements. Among organic frameworks, π -conjugated systems with electron-donating (push) and electron-accepting (pull) groups have emerged as promising NLO materials [18,19]. Their superior NLO performance stems from donor– π –acceptor (D– π –A) architectures, in which donor (D) and acceptor (A) groups are connected via π -conjugated bridges. This arrangement facilitates ICT within the molecule, leading to increased polarisability and hyperpolarisability, key mechanisms that govern nonlinear optical behaviour [18–21]. In organic D– π –A NLO materials, the first-order hyperpolarisability, driving second-order NLO responses, arises from electron transfer from the donor moiety to the acceptor moiety through the π -conjugated framework [22]. The charge transfer process enables these molecules to be used in electric-field-poled polymer-based electro-optic devices [23] and in second-harmonic generation (SHG) imaging of biological structures [21]. Despite these promising applications, identifying new organic NLO systems that combine efficient charge transfer pathways, structural stability, and large hyperpolarisability remains a significant challenge.

Triphenylphosphine oxide derivatives have emerged as promising NLO materials due to their rigid phosphorus-oxygen (P–O) framework and ability to facilitate ICT [24–30]. Literature reports that P–O-based systems can promote electronic delocalisation and intramolecular charge transfer [25,31–33]. Although the large number of crystal structures are reported in databases such as the Cambridge Structural Database (CSD) [34] and the International Union of Crystallography (IUCr) [35], many remain characterised mainly at the crystallographic level, with limited exploration of their functional properties. Diphenyl-[2-(2-pyridylaminomethyl)phenyl]phosphine oxide (OPPh₃py) is one such compound. Originally synthesised and characterised by single-crystal X-ray diffraction by one of the present authors [36], no study has yet examined the NLO properties of OPPh₃py.

This work investigates the structural, electronic, and NLO properties of OPPh₃py using an integrated experimental-theoretical approach. Density functional theory (DFT) [37] calculations are used to rationalise the experimental findings and classify the molecule based on its linear and nonlinear optical susceptibilities. Structural characterisation is discussed based on both experimental data and theoretical analysis. The ultraviolet-visible (UV-vis) spectroscopic studies, along with frontier molecular orbitals (FMOs) analysis, were used to elucidate charge transfer

characteristics. Furthermore, molecular electrostatic potential (MEP) analysis was employed to identify reactive regions. To gain further insight into the chemical reactivity of OPPh₃py, we evaluated several global reactivity parameters (GRPs), including chemical potential, electronegativity, chemical hardness, chemical softness, electrophilicity, and nucleophilicity. We also analysed local descriptors, such as Fukui functions, local softness, and philicity. Natural bond orbital (NBO) analysis was performed to explore hyperconjugative interactions and charge delocalisation. We performed DFT calculations at different levels of theory to evaluate the dipole moment and linear optical properties, including isotropic and anisotropic polarisabilities. Second-order NLO properties were evaluated from the first hyperpolarisability (β) and the second-harmonic generation (SHG) response. Third-order NLO properties were obtained from the second hyperpolarisability (γ), including the frequency-dependent component $\gamma(-2\omega; \omega, \omega, 0)$, to evaluate electric-field-induced second-harmonic generation (ESHG) responses [38,39].

Molecular structural and electronic analyses indicate that OPPh₃py is a D- π -A type organic chromophore featuring an intramolecular hydrogen bond (observed in the gas phase). NBO analysis reveals extensive hyperconjugative interactions and charge delocalisation, features commonly associated with NLO materials. The NLO responses were consistent, indicating that OPPh₃py exhibits an optical response in the off-resonance region ($\lambda > 1000$ nm). Benchmarking against *p*-nitroaniline shows that OPPh₃py exhibits superior NLO properties. Thus, the present study enables direct comparison of the NLO responses of newly designed organic compounds with those of OPPh₃py.

2. Experimental and computational methods

2.1 Experimental methods

OPPh₃py was obtained following the methodology previously reported [36]. A solution of 0.345 g 2-(diphenylphosphinyl)benzaldehyde in 5 mL of ethanol was added to an ethanolic solution containing 0.111 g of 2-aminopyridine at 20 °C. After five hours of reflux, the reaction was complete. Solvent removal yielded a light brown solid that was filtered, washed with hexane, and air dried, giving 0.408 g (89.3% yield). Further details of the proposed reaction mechanism for the synthesis of OPPh₃py are provided in Section S1, and the corresponding mechanism is shown in Scheme S1. In this study, we adopt the atomic numbering scheme shown in Figure 1. The electronic absorption spectrum in chloroform was recorded at room temperature in the 220–400 nm wavelength range using an Agilent 8453 UV-Vis spectrophotometer.

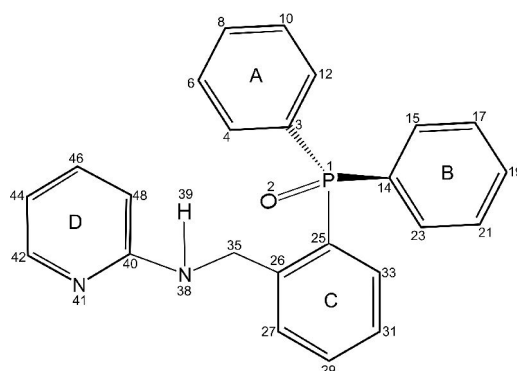


Figure 1. Molecular structure of diphenyl[2-(2-pyridylaminomethyl)phenyl]phosphine oxide (OPPh₃py).

2.2 Computational methods

All quantum computations were performed with the Gaussian 16 C.01 program [40]. Output files were analysed using various software tools, including GaussView 6.0.16 [41], Visual Molecular Dynamics (VMD) [42], and custom Python scripts. The structural parameters from the X-ray diffraction data [36] were used as the starting point for the geometry optimisation. All calculations were performed in the gas phase at room temperature and standard pressure. The geometric and electronic characteristics of OPPh₃py were determined using DFT methods along with the three-parameter hybrid functional B3 for the exchange part and the Lee-Yang-Parr (LYP) correlation functional (B3LYP), which has been extensively validated for the accurate description of structural, electronic, NBO and thermodynamic properties of organic compounds [37,43]. We carried out conformational analysis to identify the most stable conformer of OPPh₃py by performing relaxed potential energy surface (PES) scans for each relevant dihedral angle: ϕ_1 (O2-P1-C25-C26), ϕ_2 (O2-P1-C3-C4), ϕ_3 (O2-P1-C14-C15), ϕ_4 (C27-C26-C35-N38), ϕ_5 (C26-C35-N38-C40), and ϕ_6 (C35-N38-C40-N41). Each dihedral was rotated independently from 0° to 360° in 36 steps of 10° at the B3LYP/6-31+G(d,p) level of theory.

The global-minimum isomer identified in the PES scan was fully optimised at the B3LYP-D3(BJ) level, employing the 6-311++G(2df,p) basis set [44], where D3(BJ) denotes Grimme's third-generation empirical dispersion correction with Becke-Johnson damping [45–47]. The absence of imaginary frequencies confirmed the optimised structure. Moreover, time-dependent DFT (TD-DFT) [43,48] was employed to investigate the electronic properties (FMOs and GRPs) and absorption spectra of OPPh₃py. TD-DFT is widely used for predicting electronic excitation energies and UV-Vis absorption spectra with reasonable accuracy [48–50]. To take into account the effects of the solvent on the spectrum, we employed the self-consistent reaction field (SCRF) approach with the linear response polarisable continuum model (PCM) [51], with chloroform as solvent ($\epsilon = 4.9$, dipole = 1.04 D). The MEP and local reactivity parameters (charge population, Fukui function, local softness and philicity and NBO analysis were determined using the above-mentioned level of theory. NBOs were plotted using the Jmol-NBO Visualization Helper [52].

The electric dipole moment, isotropic average polarisability, polarisability anisotropy, first- and second-order hyperpolarisabilities, second-harmonic generation, and third-harmonic generation were computed in both static and dynamic regimes at excitation wavelengths $\lambda = \infty$, 1906, 1341, 1064, 625 and 532 nm. The ω B97XD (omega Becke 97X with dispersion correction) functional [53] was selected as the reference method for evaluating the NLO properties of OPPh₃py because its inclusion of long-range exchange and empirical dispersion corrections enables an accurate description of both ground and excited states [38,53]. For comparison, additional calculations were performed using the M06-2X meta-hybrid functional, B3LYP hybrid functional, and CAM-B3LYP range-separated hybrid functional [37,43] with the 6-311++G(2df,p) basis set. For functionals without an intrinsic dispersion treatment, empirical Grimme D3 or D3(BJ) dispersion corrections were applied as appropriate (More detail in [Supporting Information](#)). All calculations were performed in the gas phase (GAS) and using the PCM and the solvation model based on density (SMD) approaches [54]. In addition, a comparative assessment using the def2-TZVPPD basis set [55,56] with the M06-2X functional was performed to evaluate basis-set sensitivity. Throughout this work, the notation *M06-2X refers to the M06-2X-D3/def2-TZVPPD level of theory, whereas M06-2X denotes M06-2X-D3/6-311++G(2df,p).

3. Results and discussion

3.1 Molecular structure

3.1.1 Conformational analysis

Hernández et al. [36] reported two conformational isomers (**I** and **II**) of OPPh₃py shown in Figure 2A. We performed single-point energy calculations at the B3LYP/6-31+G(d,p) level and systematic PES scans of the dihedral angles ϕ_1 - ϕ_6 for both structures. The rotation around ϕ_5 produced the global minimum. Both isomer **I** and isomer **II** converged to nearly identical low-energy structures, labelled **Minimum I** and **Minimum II**. **Minimum I** was identified as the global minimum and was therefore taken as the reference ($\Delta E = 0$). The relative energies of all other conformers were calculated with respect to this global minimum, as shown in Figure 2B. The three-dimensional representation of the global minimum is presented in Figure 2C.

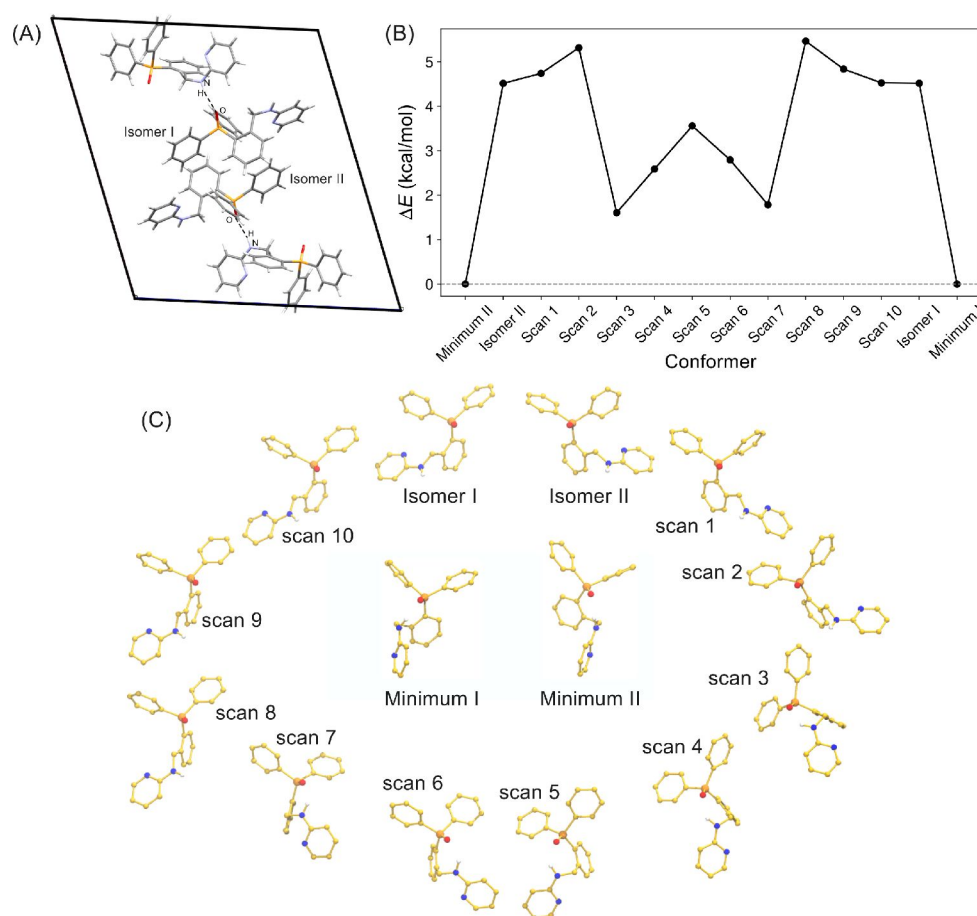


Figure 2. (A) Unit cell of OPPh₃py, highlighting Isomers I and II with inter-molecular hydrogen bonds (N-H...O). (B) Exploration of the freely rotating bond for the potential energy surface (PES) conformational analysis of OPPh₃py. The relative energy (ΔE) profile is plotted with respect to Minimum I (identical to Minimum II). (C) Three-dimensional conformer structures mapped onto the PES. Atom colours are represented as follows: C (carbon) in gold, P (phosphorus) in orange, N (nitrogen) in blue, O (oxygen) in red, and H (hydrogen) in white.

3.1.2 Geometrical parameters

Table 1 compares the selected optimised structural parameters (bond lengths, bond angles, and dihedral angles) with the corresponding experimental values. Tables S1-S2 present all the geometrical parameters for OPPh₃py. The P=O bond length in the optimised conformation (1.51

Å) is slightly elongated compared to the experimental average (1.48 Å), indicating a redistribution of electron density possibly due to intramolecular interactions in the optimised geometry. Similarly, the length of the P1-C3 bond in the optimised structure (1.83 Å) is marginally longer than the experimental values (1.80-1.81 Å), which may reflect relaxation of the steric and electronic strain during optimisation.

Table 1. Selected experimental and optimised structural parameters of OPPh₃py. The geometry optimisation was performed at the B3LYP-D3(BJ)/6-311++G(2df,p) level of theory.

Parameter	^a Experimental		optimised	Deviation (Exp _{avg} –Calc.)
	Isomer I	Isomer II		
Bond Lengths (Å)				
P1-O2	1.49 (17)	1.48 (18)	1.51	0.03
P1-C3	1.80 (2)	1.81 (3)	1.83	0.03
C8-C10	1.34 (4)	1.36 (5)	1.40	0.05
C29-C31	1.36 (4)	1.38 (4)	1.39	0.02
C35-N38	1.43 (4)	1.44 (3)	1.46	0.02
N38-C40	1.35 (4)	1.36 (3)	1.34	0.02
C42-C44	1.35 (5)	1.35 (5)	1.39	0.04
Bond Angles (°)				
C3-P1-C25	106.98 (11)	104.94 (11)	105.43	0.53
C4-C3-P1	124.10 (20)	123.60 (20)	122.51	1.34
C15-C14-P1	123.30 (20)	119.40 (20)	117.09	4.26
C27-C26-C35	120.20 (20)	121.40 (20)	118.82	1.98
N38-C35-C26	115.50 (20)	115.40 (20)	113.20	2.25
Dihedral Angles (°)				
O2-P1-C3-C4	-132.30 (20)	-40.90 (30)	-153.76	-
C25-P1-C3-C4	-167.90 (20)	-98.10 (20)	-28.88	-
C26-C35-N38-C40	-82.70 (40)	81.30 (30)	-74.07	-
Hydrogen Bond Geometry (Å,°)				
N38-H39	0.90 (10)	0.90 (1)	1.02	-
^b H39...O2	1.98 (1)	1.97 (1)	1.99	-
^c N38-H39...O2	177 (30)	166 (30)	143	-

^aNumbers in parentheses represent estimated standard uncertainties.

^{b,c}Hydrogen bond is intermolecular in experiment and intramolecular in theory.

In particular, the geometry of the N38-H39...O2 hydrogen bond reveals a significant structural change: while the experimental structures show an *intermolecular* hydrogen bond with near-linear geometry (average angle 171°, distances 1.98 Å), the optimised structure features a longer *intramolecular* hydrogen bond (1.99 Å) but with a much smaller angle of 143°, indicating reduced directionality. This intramolecular hydrogen bonding likely stabilises a folded geometry, contributing to the observed deviations in dihedral angles (e.g., C27-C26-C35-N38 and C25-C26-C35-N38) and promoting a more polarised electronic distribution across the molecular backbone. The presence of a strong intramolecular hydrogen bond between the donor (NH) and acceptor (P=O) groups supports intramolecular charge transfer by facilitating electron delocalisation across the conjugated framework (as further discussed in sections [Electronic structure](#) and [Natural bond orbital analysis](#)). This internal proton bridge effectively connects electron-rich and electron-deficient parts of the molecule, which is a key requirement for enhanced first-order hyperpolarisability, a key indicator of NLO activity [16]. Furthermore, the moderate deviation in the N38-C35-C26 bond angle (optimised: 113.20° *vs* experimental average: 115.50°) suggests increased torsional flexibility. This flexibility allows the molecule to adopt conformations that favour dipole moment changes under external fields, which contributes to the charge transfer-mediated NLO response [57].

PES analysis identified the intramolecularly hydrogen-bonded conformer as the gas-phase global minimum. However, this geometry is not observed in the experimentally determined crystal structure, where intermolecular hydrogen bonds dominate because of lattice-packing effects. This discrepancy indicates that crystal packing stabilises alternative intermolecular interactions and suppresses the formation of the folded, intramolecularly hydrogen-bonded geometry. Therefore, for solid-state NLO applications, stabilisation of this conformation would require control of the intermolecular packing, potentially through polymorph selection, substitution strategies, or incorporation into amorphous or polymeric matrices [58,59].

The comparison in Table 1 highlights the dual structural and electronic role of intramolecular hydrogen bonding. It reinforces the charge transfer character of the molecule, which is central to its NLO properties. Moreover, as shown in Figure 2A, OPPh₃py exhibits a non-centrosymmetric structure, lacking an inversion center, an essential symmetry criterion for second-order NLO materials [57].

3.2 Electronic structure

3.2.1 Molecular electrostatic potential

The MEP mapping technique is used to predict the electrophilic and nucleophilic regions of OPPh₃py, where the negatively charged regions of the molecule are coloured red and the positively charged regions are coloured blue. Phenyls A and B exhibit the most positive electrostatic regions (Figure 3). The region with the highest negative charge (related to electrophilic reactivity) is located near the O2 atom due to the presence of its lone pair electrons. The N38-H39 and pyridyl groups are characterised by regions of negative electrostatic potential. These regions are susceptible to electrophilic attack. Therefore, this analysis confirms the presence of an intramolecular interaction between the O2 and N38-H39 groups in OPPh₃py, as observed in the PES scan.

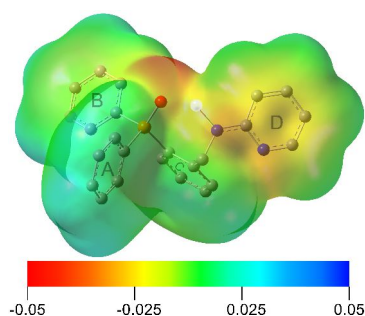


Figure 3. MEP map for OPPh₃py (values in a.u.). The blue and red colours represent the positive and negative iso-potential, respectively.

3.2.2 Frontier molecular orbitals

The FMO analysis provides valuable insight into a molecule's electronic structure, chemical reactivity, and optical properties [60,61]. The HOMO and LUMO energies, along with the energy gap ($E_{\text{LUMO}} - E_{\text{HOMO}}$), were calculated for OPPh₃py and compared with those of phosphine oxide derivatives [62–65], as illustrated in Table 2 and Table S3. Table 2 shows that OPPh₃py exhibits a moderate HOMO energy (-5.44 eV) and a relatively high LUMO energy (-1.37 eV), resulting in a HOMO-LUMO energy gap of 4.07 eV. This energy gap is larger than those reported for the reference phosphine oxides TPP2 (3.07 eV) and compound TPP1 (3.20 eV), but is

comparable to the values of **TPP4** (4.05 eV) and **TPP3** (4.42 eV), which are known for their chemical reactivity. These results indicate that **OPPh₃py** shows lower electronic reactivity and greater kinetic stability compared to more reactive analogues with smaller energy gaps. Beyond the HOMO-LUMO energy gap, the ability of compounds to undergo intramolecular charge transfer from donor to acceptor groups via π -conjugated linkers is a key factor in NLO structure-function relationships [57,66]. Figure 4 shows the contour surfaces of the FMOs, illustrating charge distribution and transfer.

The HOMO is predominantly located on the donor pyridyl (ring D) and amino groups. The LUMO density is distributed over the acceptor phenyl rings A and B, with a minor contribution from phenyl ring C, which exhibits π -character (as further discussed in the [Natural bond orbital analysis](#) section). Moreover, the electron donor and acceptor are linked via a π -conjugation bridge (phenyl ring C), which facilitates intramolecular charge transfer. Although **OPPh₃py** has a moderate HOMO–LUMO energy gap, the molecular structure features strong electron-donating and electron-withdrawing groups separated by a π -delocalised moiety. This is further supported by the presence of an intramolecular hydrogen bond involving N38–H39 and the O2 moiety. Therefore, this charge transfer behaviour suggests the potential of this molecule for use in nonlinear optical applications.

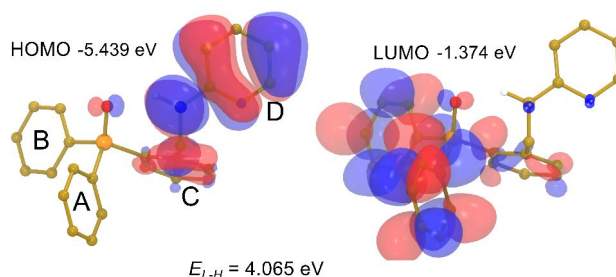


Figure 4. HOMO and LUMO orbitals for **OPPh₃py**. The blue colour denotes the positive isosurface, while the red colouration represents the negative isosurface. All values are expressed in electron volts (eV).

3.2.3 Global chemical parameters

The GRPs were calculated for **OPPh₃py** to assess its reactivity and stability. For benchmarking purposes, the results were compared with those of reference phosphine oxides reported in the literature [62–65]. Table 2 summarises the GRPs values. The energies of the FMOs (E_{HOMO} and E_{LUMO}) provide insight into global reactivity parameters, including the ionisation potential (IP) and electron affinity (EA), which can be estimated using Koopmans' theorem [67]:

$$IP = -E_{\text{HOMO}} \quad (3.1)$$

$$EA = -E_{\text{LUMO}} \quad (3.2)$$

where E_{HOMO} is the HOMO energy and E_{LUMO} is the LUMO energy. The chemical potential (μ), electronegativity (χ), global hardness (η), global softness (S), global nucleophilicity index (ω^-), global electrophilicity index (ω^+) can be computed using the following Eqs. (3.3–3.8) below; [67,68]

$$\mu = -\frac{IP + EA}{2} \quad (3.3)$$

$$\chi = -\mu \quad (3.4)$$

$$\eta = \frac{IP - EA}{2} \quad (3.5)$$

$$S = \frac{1}{\eta} \quad (3.6)$$

$$\omega^- = \frac{(3IP + EA)^2}{16(IP - EA)} \quad (3.7)$$

$$\omega^+ = \frac{(IP + 3EA)^2}{16(IP - EA)} \quad (3.8)$$

Table 2. Global reactivity parameters (E_H , E_L , E_{L-H} : energy of FMO, μ : chemical potential, χ : electronegativity, η : global hardness, global softness, ω^- : nucleophilicity index, ω^+ : electrophilicity index) for OPPh₃py and reference molecules. All values are in eV.

^a Molecule	E_H	E_L	E_{L-H}	μ	χ	η	S	ω^+	ω^-
OPPh ₃ py	-5.44	-1.37	4.07	-3.41	3.41	2.04	0.49	0.71	4.81
TPP1 [64]	-5.64	-2.44	3.20	-4.04	4.04	1.60	0.63	1.28	7.32
TPP2 [63]	-5.22	-2.15	3.07	-3.68	3.68	1.53	0.65	1.11	6.46
TPP3 [62]	-5.45	-1.03	4.42	-3.24	3.24	2.21	0.45	0.59	4.27
TPP4 [65]	-5.65	-1.60	4.05	-3.62	3.62	2.03	0.49	0.81	5.31
TPP5 [65]	-5.52	-1.67	3.85	-3.59	3.59	1.92	0.52	0.84	5.40

^aLiterature values were calculated using the B3LYP method in conjunction with the 6-31G(d) basis set.

The GRPs in Table 2 position OPPh₃py as a moderately reactive and electronically balanced phosphine oxide. The molecule shows a HOMO-LUMO gap of 4.07 eV, comparable to **TPP3** (4.42 eV) and larger than the highly reactive reference compound (**TPP1**, 3.20 eV). This relatively wide gap indicates good electronic stability, reducing the likelihood of undesired charge transfer events under operational conditions. OPPh₃py exhibits a chemical potential of -3.41 eV and an electronegativity of 3.41 eV, values that place the molecule between the most reactive analogues **TPP2** and **TPP1** and the more stable **TPP3**. This intermediate profile implies controlled electron-accepting and electron-donating behaviour, supporting efficient yet non-excessive polarisation of the electronic cloud. The molecule's global hardness (2.04 eV) and softness (0.49 eV) further confirm its position as a structurally robust yet sufficiently polarisable system. Compared with highly reactive phosphine oxides such as compound **TPP1**, OPPh₃py remains less susceptible to chemical degradation, which is advantageous for device-oriented NLO applications [16]. OPPh₃py also displays moderate electrophilicity ($\omega^+ \approx 0.71$ eV) and nucleophilicity ($\omega^- \approx 4.81$ eV). These values are lower than those of strongly reactive analogues (e.g., **TPP1** and **TPP2**), demonstrating that OPPh₃py does not excessively favour electron donation or acceptance. This balanced profile contributes to operational durability, particularly in environments where extreme reactivity can compromise material lifetime.

The GRPs analysis indicates that OPPh₃py is highly stable and exhibits notable nucleophilicity and softness. This favourable reactivity under nucleophilic conditions supports intramolecular charge transfer between electron-donating and electron-accepting moieties through conjugated pathways, a key mechanism in nonlinear optical behaviour [18,69].

3.2.4 Local chemical parameters

The condensed Fukui functions provide a means to evaluate the reactivity of individual atoms within a molecule. The corresponding expressions for electrophilic and nucleophilic attacks are given by:

$$f_k^- = q_k(N-1) - q_k(N) \quad (3.9)$$

$$f_k^+ = q_k(N) - q_k(N+1) \quad (3.10)$$

For radical attacks, the Fukui function is defined as:

$$f_k^0 = [q_k(N+1) - q_k(N-1)]/2 \quad (3.11)$$

where q_k represents the gross charge of atom k in the molecule [70]. The total number of electrons in the anionic, cationic, and neutral states of the molecule corresponds to $N+1$, $N-1$ and N , respectively. At long intermolecular distances, electrostatic interactions dominate over orbital interactions. In such cases, alternative reactivity descriptors, such as local softness ($s_k^\pm = S f_k^\pm$) [71] and philicity ($\omega_k^\pm = \omega f_k^\pm$), [71] may be more suitable. Here the + and - signs denote nucleophilic and electrophilic character, respectively. Table S4 presents the Fukui functions, local softness, and philicity values determined using CM5 [72] population analysis for OPPh₃py.

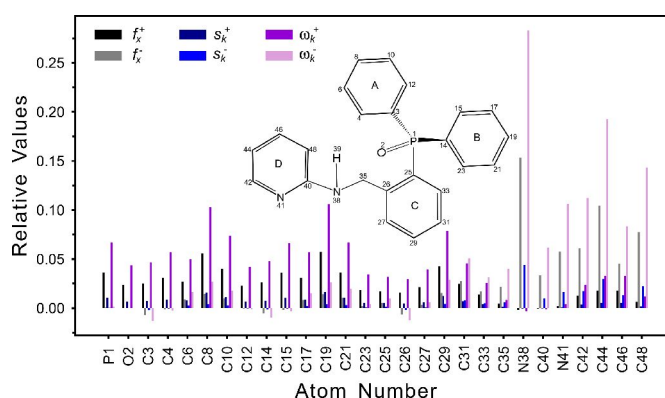


Figure 5. Fukui functions using CM5 population analysis, local softness ($s_k^\pm$) and philicity ($\omega_k^\pm$) in eV for selected atomic sites of OPPh₃py.

Based on the nucleophilic Fukui function (f_k^+), local softness (s_k^+), and philicity (ω_k^+), the most nucleophilic carbon atoms are C19 ($f_k^+ = 0.0574$, $s_k^+ = 0.0164$, $\omega_k^+ = 0.1059$), C8 ($f_k^+ = 0.0558$, $s_k^+ = 0.0160$, $\omega_k^+ = 0.1059$) and C29 ($f_k^+ = 0.0427$, $s_k^+ = 0.0122$, $\omega_k^+ = 0.0787$). These atoms are expected to be the most reactive toward electrophilic attacks (Figure 5). For nucleophilic reactivity, the most reactive site is N38 ($f_k^- = 0.1534$, $s_k^- = 0.0439$, $\omega_k^- = 0.2829$). Other important nucleophilic sites include C44 ($f_k^- = 0.1046$, $s_k^- = 0.0299$, $\omega_k^- = 0.1929$) and C42 ($f_k^- = 0.0609$, $s_k^- = 0.0174$, $\omega_k^- = 0.1123$). These atoms are likely to engage in interactions with electrophiles due to their high negative philicity values, particularly nitrogen N38. In contrast, carbon atoms such as C3, C4, C14, and C26 exhibit small or negative f_k^- , s_k^- , and ω_k^- , indicating limited nucleophilic reactivity. This is consistent with their locations in less reactive or sterically hindered regions. In contrast, C6, C17, and C21 display moderate electrophilic reactivity, suggesting they could act as electron acceptors in aromatic π -systems or conjugated frameworks. The phosphorus atom (P1) exhibits moderate nucleophilic character ($f_k^+ = 0.0364$, $s_k^+ = 0.0104$, $\omega_k^+ = 0.0671$), while its electrophilic reactivity is negligible, aligning with its role in bond polarisation. The oxygen atom (O2) displays low to moderate reactivity, with a slight preference for nucleophilic behaviour ($f_k^+ = 0.0237$, $s_k^+ = 0.0068$, $\omega_k^+ = 0.0437$).

Broadly speaking, the molecule exhibits localised regions of high reactivity, particularly around specific carbon atoms and the nitrogen atom (N38). In contrast, phosphorus and oxygen atoms display moderate to low reactivity. This, along with the observed donor- π -acceptor structure, supports efficient charge delocalisation. Combined with its global softness and nucleophilicity, these features confirm the molecule's potential as a NLO candidate.

3.2.5 UV-Vis spectrum and electronic transitions

UV-Vis absorption spectra for singlet-singlet transitions were computed using the optimised ground-state geometry to analyse the spectral features and electronic transitions of the title compound. The experimental UV-Vis spectrum was compared with the one obtained from TD-DFT-D3(BJ)/B3LYP calculations using the 6-311++G(2df,p) basis set. The solvent effect of chloroform was accounted for through the PCM model, which treats the solvent as a continuous dielectric medium surrounding the solute. Table 3 lists the calculated vertical electronic excitation energies, their oscillator strengths (f), and the corresponding assignments in the solvent phase. Figure 6 shows the experimental and simulated absorption spectra of OPPh₃py.

Table 3. Experimental and DFT calculated UV-Vis absorption parameters of OPPh₃py: wavelengths (λ , nm), excitation energies (E , eV), oscillator strengths (f), transition types, and principal molecular orbital (MO) contributions.

Transition	Exp. λ (nm)	Calc. λ (nm)	E (eV)	f	Type	b MO Contributions (%)
$S_0 \rightarrow S_{22}$	246	243	5.098	0.093	$\pi \rightarrow \pi^*$	H $\rightarrow$ L+8 (24.43), H-1 $\rightarrow$ L+5 (18.89), H-2 $\rightarrow$ L (7.67)
$S_0 \rightarrow S_{23}$			5.119	0.186		H $\rightarrow$ L+8 (22.29), H-1 $\rightarrow$ L (13.65), H $\rightarrow$ L+7 (9.75)
$S_0 \rightarrow S_{19}$	264	268	4.624	0.023	$\pi \rightarrow \pi^*$	H $\rightarrow$ L+6 (42.82), H $\rightarrow$ L+3 (10.02), H $\rightarrow$ L+4 (6.78)
$S_0 \rightarrow S_{15}$	284	286	4.340	0.048	$\pi \rightarrow \pi^*$	H $\rightarrow$ L+5 (45.27), H $\rightarrow$ L+4 (9.12)
$S_0 \rightarrow S_{13}$	327	296	4.199	0.021	$n \rightarrow \pi^*$	H $\rightarrow$ L+3 (45.81), H $\rightarrow$ L+6 (6.17)
$S_0 \rightarrow S_9$		315	3.945	0.002		H $\rightarrow$ L (29.78), H $\rightarrow$ L+2 (23.91), H-3 $\rightarrow$ L+3 (16.44)
$S_0 \rightarrow S_6$		326	3.795	0.018		H $\rightarrow$ L+1 (29.50), H-1 $\rightarrow$ L+5 (24.48), H-2 $\rightarrow$ L+2 (15.92)

^a Transitions selected for $f > 0.001$

^b H: HOMO, L: LUMO.

The experimental UV-Vis spectrum (Figure 6) reveals an intense absorption peak at 246 nm, closely matching the theoretical λ_{max} value of 243 nm. This transition exhibits the highest energy (5.12 eV) and the largest oscillator strength (0.186), making it the most probable electronic excitation in the molecule. The excitation predominantly originates from the HOMO to LUMO+8 transition, which contributes 46.72% ($S_0 \rightarrow S_{22} + S_0 \rightarrow S_{23}$) to the overall process (Table 3).

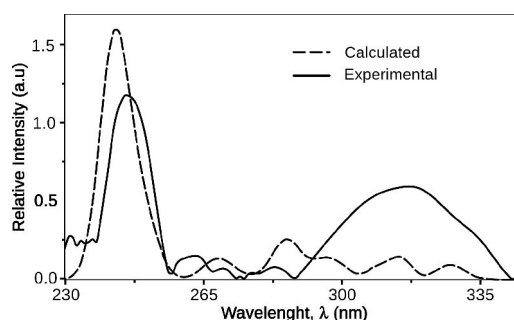


Figure 6. Experimental (solid line) and calculated (dash line) UV-Vis spectrum for OPPh₃py both are in chloroform solvent.

Analysis of molecular orbital coefficients reveals that the atoms involved in the major transitions primarily contribute through p -orbitals. Based on the computed orbital coefficients

and the molecular orbital plots of the title compound, primary electronic transitions are classified as $\pi \rightarrow \pi^*$, with evident intramolecular charge transfer from the donor to the acceptor fragment. Additional absorption bands at 264 and 284 nm in the experimental spectrum align well with the simulated peaks at 268 and 286 nm, respectively. These transitions have similar energy magnitudes and oscillator strengths of 0.023 and 0.048, respectively. The absorption band at 268 nm in the theoretical spectrum corresponds to the HOMO $\rightarrow$ LUMO+6 transition, contributing 42.82%. The broad experimental absorption band observed at 327 nm corresponds to an aggregate of three electronic excitations calculated at 296, 315 and 326 nm, with comparable oscillator strengths. TD-DFT calculations indicate that this lower-energy absorption band is primarily due to vertical excitations, involving HOMO $\rightarrow$ LUMO (45.81% in $S_0 \rightarrow S_{13}$ and 29.78% in $S_0 \rightarrow S_9$) and HOMO $\rightarrow$ LUMO+1 (29.50% in $S_0 \rightarrow S_6$), assigned to $n \rightarrow \pi^*$ charge transfer from the pyridyl and amino groups to the triphenylphosphine oxide moiety. This result is in close agreement with the findings of Duan et al. [73], who studied derivatives of triphenylphosphine oxide.

The UV-Vis spectral analysis of OPPh₃py indicates effective intramolecular charge transfer through transitions π to π^* and n to π^* . The involvement of donor and acceptor units, confirmed by orbital contributions and consistent theoretical and experimental data, suggests a favourable electronic structure for nonlinear optical applications. The calculated electronic properties reveal a well-defined intramolecular charge transfer pathway. Also, the confinement of the absorption bands of OPPh₃py to the UV region (200–400 nm), together with its pronounced nonlinear optical response at wavelengths above 1000 nm (more details on this will be given in the [Molecular dipole and optical response](#) section), indicates a favourable balance between optical transparency and nonlinear efficiency [74]. Such off-resonant NLO behaviour is highly desirable because it reduces absorption-related losses and photothermal effects while preserving strong electronic polarisation arising from intramolecular charge transfer [75].

3.3 Natural bond orbital analysis

NBO analysis was performed using second-order perturbation theory to characterise donor-acceptor interactions through examination of the Fock matrix. This study systematically explored the interactions between filled (donor) Lewis-type NBOs and empty (acceptor) non-Lewis NBOs, quantifying their energetic contributions through second-order perturbation theory. The stabilisation energy, $E(2)$, was calculated using the expression: [76]

$$E(2) = \Delta E_{ij} = q_i \left[\frac{(F_{ij})^2}{E_j - E_i} \right] \quad (3.12)$$

where $E(2)$ represents the stabilisation energy, q_i is the donor orbital occupancy, F_{ij} is the off-diagonal Fock matrix element, and E_i and E_j are the diagonal elements corresponding to the donor and acceptor orbital energies. A larger $E(2)$ value indicates a stronger interaction, typically arising from a smaller ΔE_{ij} (energy gap) or a larger F_{ij} (better orbital symmetry between the donor and acceptor) [76]. Table 4 presents selected NBO analysis results for OPPh₃py, obtained using the NBO 6.0 software [52,76,77] with the chosen functional. The complete list of electronic transitions appears in Table S5.

Two primary types of electronic transitions were observed: LP $\rightarrow \pi^*$ (lone pair to antibonding interactions) and $\pi \rightarrow \pi^*$ (bonding to antibonding interactions). The overlap between bonding and antibonding orbitals gives rise to intramolecular interactions. The strongest lone pair-to-antibonding interaction occurs between the lone pair on N38 and the π^* orbital of the C40–N41 bond, with a stabilisation energy of 53.84 kcal/mol. Additional notable interactions include LP O2 $\rightarrow \pi^*$ P1–C14, which increases the electron density (ED) by 0.12 e, and LP N41 $\rightarrow \pi^*$ C40–C48, which increases the ED by 0.03 e, leading to stabilisation energy values of 17.75 kcal/mol and 9.68 kcal/mol, respectively. The weakest interaction is observed between LP O2 and the antibonding σ^* orbital of the N38–H39 bond, with a stabilisation energy of 2.95

1 kcal/mol. These lone pairs serve as strong electron donors, contributing to hyperconjugative
2 interactions with both σ^* and π^* orbitals (Figure 7).

Table 4. Selected second-order perturbation theory analysis of Fock matrix in NBO basis for OPPh₃py.

Donor (<i>i</i>)	Type	^a ED(<i>i</i>) (e)	Acceptor (<i>j</i>)	ED(<i>j</i>) (e)	Type	^b <i>E</i> (2) (kcal/mol)	^c <i>E</i> (<i>j</i>) – <i>E</i> (<i>i</i>) (a.u)	^d <i>F</i> (<i>i</i> , <i>j</i>) (a.u)
LP O2	-	1.976	BD* N38-H39	0.028	σ^*	2.95	1.03	0.049
LP O2	-	1.833	BD* P1-C14	0.118	π^*	17.75	0.5	0.084
LP N38	-	1.743	BD* C40-N41	0.460	π^*	53.84	0.26	0.105
LP N41	-	1.910	BD* C40-C48	0.032	π^*	9.68	0.83	0.08
BD C3-C4	π	1.681	BD* C6-C8	0.322	π^*	18.13	0.29	0.065
BD C6-C8	π	1.650	BD* C10-C12	0.303	π^*	19.32	0.28	0.066
BD C26-C27	π	1.629	BD* C25-C33	0.373	π^*	20.03	0.26	0.065
BD C40-N41	π	1.710	BD* C42-C44	0.311	π^*	30.15	0.32	0.088

^a electron density (ED); ^b *E*(2) means energy of hyperconjugative interactions; ^c energy difference between donor (*i*) and acceptor (*j*). ^d *F*(*i*, *j*) is the Fock matrix element between *i* and *j* NBOs

3 The strong LP $\rightarrow \pi^*$ interactions, particularly the donation from N38 to the C40-N41 π^*
4 orbital (53.84 kcal/mol), promote intramolecular charge delocalisation. This delocalisation
5 increases the electronic polarisability of OPPh₃py. These hyperconjugative effects, along with
6 additional LP contributions from O2 and N41, support the molecule's potential for nonlinear
7 optical applications by facilitating charge redistribution along its conjugated framework.
8 Notably, the HOMO is also localised in this region, further highlighting its role as the principal
9 electron-donating site in charge transfer processes that drive NLO activity.

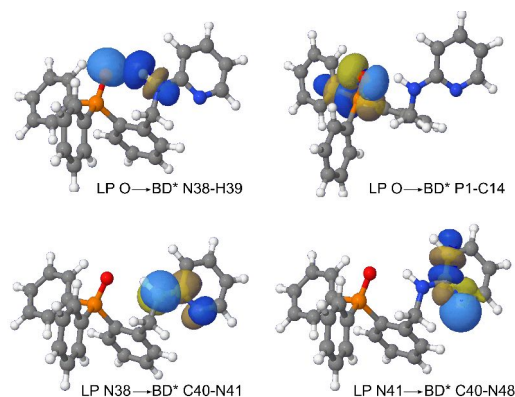


Figure 7. Electron donating NBOs interactions type LP in OPPh₃py.

10 3.3.1 $\pi \rightarrow \pi^*$ interactions and molecular stability

11 Numerous $\pi \rightarrow \pi^*$ interactions occur between various carbon-carbon bonds, with *E*(2) values
12 ranging from 13.34 to 30.15 kcal/mol. These interactions enhance electron delocalisation, thereby
13 increasing both molecular stability and reactivity. Notable interactions include
14 $\pi(\text{C3} - \text{C4}) \rightarrow \pi^*(\text{C6} - \text{C8})$ and $\pi(\text{C6} - \text{C8}) \rightarrow \pi^*(\text{C10} - \text{C12})$, with stabilisation energies of
15 18.32 and 19.32 kcal mol⁻¹, respectively. Notably, the acceptor orbitals involved in these
16 interactions contributes to the LUMO, as evidenced by its spatial localisation over these same
17 carbon atoms. This correspondence reinforces the view that the LUMO is delocalised over the
18 most conjugated segments of the molecule, consistent with the observed NBO donor-acceptor

patterns. The observed $\pi \rightarrow \pi^*$ interactions, such as $\pi(\text{C26} - \text{C27}) \rightarrow \pi^*(\text{C25} - \text{C33})$, $\pi(\text{C25} - \text{C33}) \rightarrow \pi^*(\text{C26} - \text{C27})$, $\pi(\text{C25} - \text{C33}) \rightarrow \pi^*(\text{C29} - \text{C31})$ highlight the delocalised electron flow through the phenyl ring C (Figure 8). These interactions, along with the HOMO-LUMO distribution localised over the donor and acceptor regions, confirm the role of the central phenyl carbon atoms as effective π -linkers. The acceptor orbitals, particularly the π^* antibonding orbitals of bonds such as P1-C14 and P1-C3 play a crucial role in determining molecular reactivity. These orbitals attract electron density from lone pairs and π -bonds, forming reactive sites for electrophilic attack.

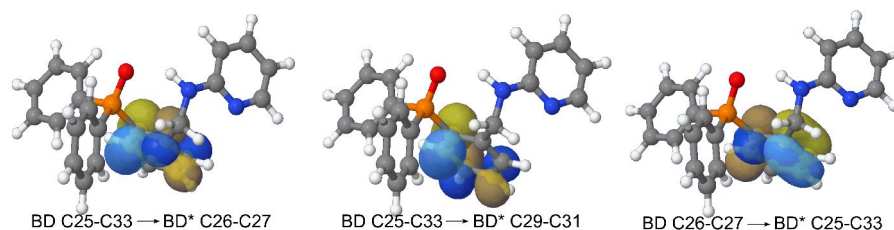


Figure 8. NBO representation of the delocalised electron flow through the phenyl ring C.

3.3.2 Canonical molecular orbital analysis

The simulated UV-Vis spectrum was utilised to perform canonical molecular orbital (CMO) analysis, providing insights into delocalised molecular orbitals by decomposing them into their leading natural bond orbital contributions (bonding, nonbonding, or antibonding). This approach offers a concise NBO-based description of each molecular orbital [52,76,78]. Within the NBO framework, a sequence of localised transformations is applied instead of a direct atomic orbital (AO) $\rightarrow$ molecular orbital transformation:

$$\text{AO} \rightarrow \text{NAO} \rightarrow \text{NBO} \rightarrow \text{NLMO} \rightarrow \text{CMO}$$

where AO, NAO, NBO, NLMO, and CMO represent atomic orbitals, natural atomic orbitals, natural bond orbitals, natural localised molecular orbitals, and canonical molecular orbitals, respectively. Each molecular orbital (ψ_i) can be expressed as a linear combination of NBOs (Ω_j) using:

$$\psi_i^{\text{CMO}} = \sum_j \Omega_{ij} \cdot t_{ij}^{(\text{NBO})} \quad (3.13)$$

where t_{ij} is the weighting coefficient that determines the percentage contribution of each NBO to the CMO expansion of ψ_i . Table 5 presents the NBO contributions to the HOMO and LUMO, computed using the NBO 6.0 software. Table 5 indicates that the HOMO cluster is an out-of-phase mixture, primarily composed of the lone pair (LP) on N38 (37.21%), followed by three π -bonding contributions from C42-C44 (24.80%), C40-N41 (12.46%) and C46-C48 (9.36%). Additionally, a small π^* -antibonding contribution from C46-C48 (7.24%) is observed. The LUMO cluster is composed of six NBOs, predominantly contributing from π^* -antibonding orbitals (Figure 9). This delocalisation over regions A and B agrees with the distribution shown in Figure 4. The HOMO exhibits a dominant contribution from LP N38 and the π -bonding NBO of C42-C44, whereas the LUMO is primarily composed of the π^* antibonding orbitals associated with C14-C15, C3-C4, C17-C19 and C6-C8.

The NBO analysis enhances our understanding of the relationship between the electronic structure and the NLO performance of OPPh₃py. The molecular reactivity is primarily influenced by nitrogen-containing regions and the phosphorus center, while the aromatic

Table 5. Details on the selected canonical molecular orbitals (HOMO and LUMO) of OPPh₃py calculated from the B3LYP density matrix.

CMO cluster (ψ_i)	CMO (Ω_j)	Weighting Factor (t_{ij})	^a %Contribution
HOMO	LP N38	0.610	37.21
	π C42-C44	-0.498	24.80
	π C40-N41	0.353	12.46
	π C46-C48	0.306	9.36
	π^* C46-C48	0.269	7.24
LUMO	π^* C14-C15	-0.443	19.61
	π^* C3-C4	0.439	19.27
	π^* C17-C19	-0.360	12.96
	π C21-C23	-0.284	8.06
	π^* C6-C8	0.282	7.95
	π C10-C12	0.266	7.08

^aCalculated as: $100 \cdot |t_{ij}|^2$

1 systems contribute to electron delocalisation, which is crucial for enhancing NLO properties.
 2 High $E(2)$ values, particularly those corresponding to interactions between LP and π^* orbitals,
 3 or between π and π^* orbitals, indicate strong electron delocalisation, a key characteristic for
 4 nonlinear optical activity [79–81]. The strong donor-acceptor interactions observed within the
 5 π -conjugated framework indicate effective charge transfer across the molecular backbone. Such
 6 charge redistribution promotes electronic delocalisation and is closely linked to the enhancement
 7 of the first- and second-order hyperpolarisabilities [61]. These findings demonstrate that
 8 intramolecular charge transfer plays a central role in the favourable NLO behaviour of OPPh₃py.

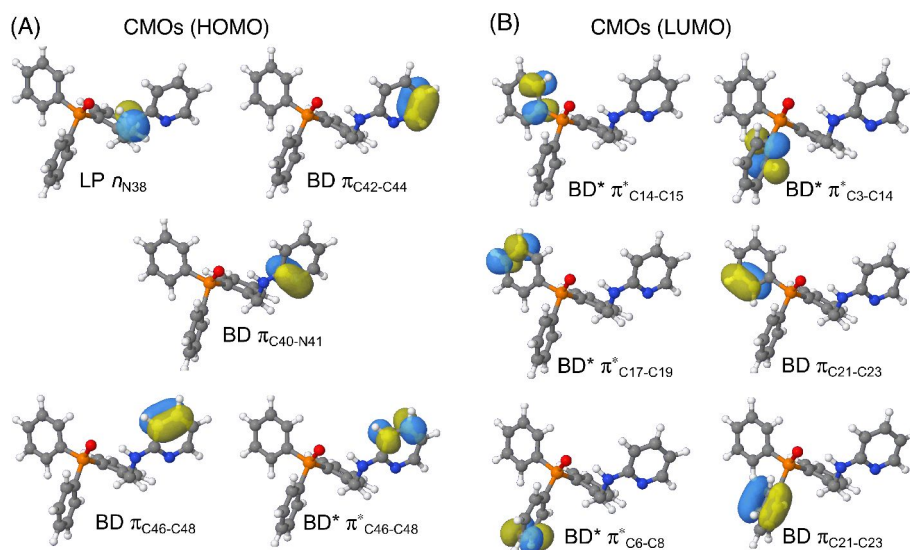


Figure 9. Visualisation of (A) HOMO and (B) LUMO clusters from canonical molecular orbital calculations of OPPh₃py using the NBO method.

3.4 Molecular dipole and optical response

In relation to the electronic transitions discussed in Section 3.3.5, frequency-dependent dispersion analysis was conducted to evaluate the dynamic nonlinear optical response at representative Nd:YAG laser excitation wavelengths of ∞ , 1906, 1341, 1064, 625 and 532 corresponding respectively to photon energies (ω) of 0.0, 0.024, 0.034, 0.043, 0.074, and 0.085 a.u., together with the static limit at $\omega = 0$. However, the present study does not discuss the results obtained at 625 and 532 nm because the measurements showed substantial deviations and anomalous behaviour under resonance conditions. The complete dataset is available in the online repository [82]. The substantial deviations observed in the resonance region ($\lambda < 1000$ nm) may arise from the predominance of nonresonant nonlinear optical behaviour in OPPh₃py. In this section, *p*-nitroaniline (*p*-NA) was employed as a reference compound because experimental standards are not available for OPPh₃py. The reference values for *p*-NA correspond to the static field limit ($\omega = 0$) and are given as follows: $\mu_{D(p\text{-NA})} = 6.06$ Debye, $\alpha_{\text{iso}(p\text{-NA})} = 13.8 \times 10^{-24}$ esu, $\alpha_{\text{aniso}(p\text{-NA})} = 7.5 \times 10^{-24}$ esu, $\beta_{(p\text{-NA})} = 0.6 \times 10^{-30}$ esu, and $\gamma_{(p\text{-NA})} = 7.5 \times 10^{-36}$ esu. These values were reported at the $\omega\text{B97XD}/6\text{-311++G(d,p)}$ level of theory [83].

3.4.1 Dipole moment, linear isotropic and anisotropic polarisabilities

The dipole moment (μ_D) is an electronic property arising from the nonuniform distribution of electronic charge over the molecular framework. Its magnitude provides valuable information regarding molecular geometry, charge separation, and intramolecular electronic delocalisation [57,84]. In donor- π -acceptor systems, an enhanced dipole moment generally promotes stronger NLO activity due to the increased efficiency of intramolecular charge transfer [57]. Another important electronic property closely associated with charge transfer efficiency and optical response is molecular polarisability. The isotropic and anisotropic polarisabilities are fundamental linear optical parameters that describe the response of a molecular electron cloud to an external electric field. The isotropic polarisability (α_{iso}) represents the average electronic polarisation over all spatial directions, whereas the anisotropic polarisability (α_{aniso}) measures the directional dependence of this response and reflects the degree of electronic delocalisation and molecular asymmetry. Molecules possessing extended π -conjugation and strong donor-acceptor interactions generally exhibit enhanced polarisability values due to the increased mobility of π -electrons under an applied electric field [16]. The magnitudes of the dipole moment and the isotropic and anisotropic polarisabilities were determined using Eqs. 3.14–3.16, respectively [85]. The corresponding values are reported in Table S6, while the calculated trends at different levels of theory for OPPh₃py are illustrated in Figure 10.

$$\mu_D = \left(\mu_x^2 + \mu_y^2 + \mu_z^2 \right)^{1/2} \quad (3.14)$$

$$\alpha_{\text{iso}} = \frac{1}{3} (\alpha_{xx} + \alpha_{yy} + \alpha_{zz}) \quad (3.15)$$

$$\alpha_{\text{aniso}} = \frac{1}{\sqrt{2}} \left[(\alpha_{xx} - \alpha_{yy})^2 + (\alpha_{xx} - \alpha_{zz})^2 + (\alpha_{yy} - \alpha_{zz})^2 \right]^{1/2} \quad (3.16)$$

For the dipole moment, gas-phase values remain nearly constant across wavelengths, ranging from 4.57 to 4.80 D depending on the functional (Figure 10A). Relative to the B3LYP reference, CAM-B3LYP and M06-2X predict slightly smaller dipole moments, typically about 1.01–1.05 times lower, while ωB97XD yields values essentially equivalent to B3LYP (within a factor of 0.99–1.00). In contrast, solvent effects significantly increase the dipole moment. Within the PCM model, μ_D increases from 4.80 D in the gas phase to 6.13 D, corresponding to values that are approximately 1.28–1.35 times larger. In the SMD model, the enhancement is slightly stronger, reaching 6.33 D, which is about 1.32–1.40 times larger than the gas-phase reference. The arrow in

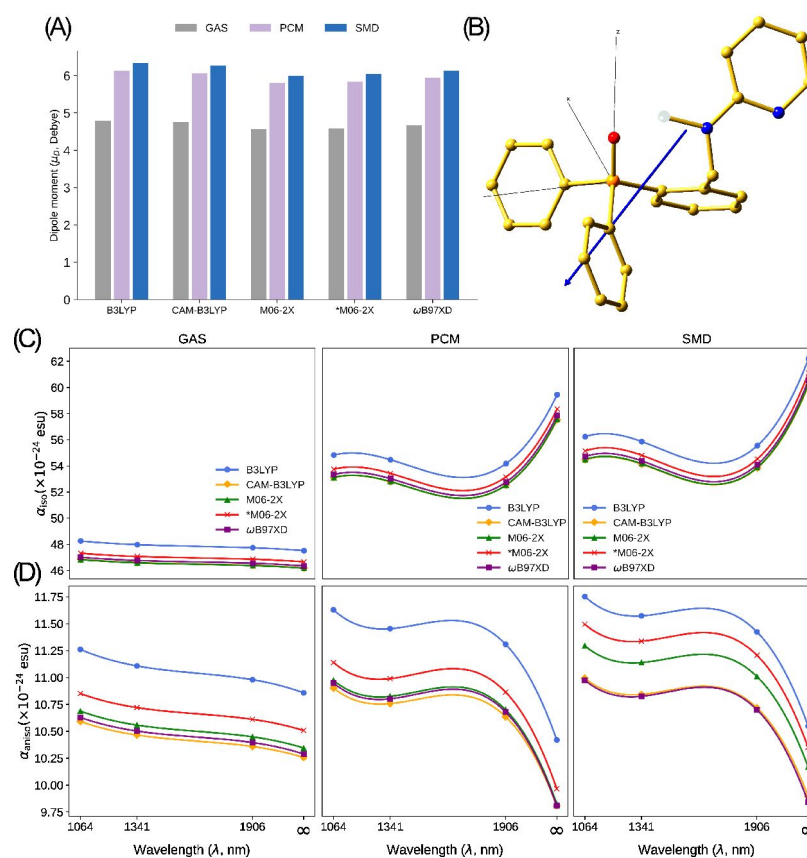


Figure 10. (A) Dipole moments of the neutral OPPh₃py molecule calculated using different levels of theory. (B) Blue arrows indicate the orientation of the dipole moment vectors. Wavelength-dependent isotropic (C) and anisotropic (D) polarisabilities obtained using the GAS, PCM, and SMD solvation models.

1 **Figure 10B** indicates the direction of the ground-state dipole moment. The dipole moment is
 2 oriented from the pyridyl ring (donor group) across the phenyl groups (π -bridges), creating a
 3 polarisable electronic distribution, consistent with the MEP map shown in **Figure 3**.

4 The calculated isotropic polarisability (α_{iso}) of OPPh₃py depends strongly on the density
 5 functional and solvent environment (**Figure 10C**), whereas its frequency dependence remains
 6 modest within the investigated nonresonant region. In the gas-phase static limit, α_{iso} ranges
 7 from 46.17 to 47.51×10^{-24} esu, with inter-functional deviations within 3%, indicating good
 8 methodological consistency. Compared with *p*-nitroaniline (*p*-NA; $\alpha_{\text{iso}} = 13.8 \times 10^{-24}$ esu),
 9 OPPh₃py exhibits isotropic polarisabilities approximately 3.3–3.4 times larger, reflecting greater
 10 electronic delocalisation and enhanced optical responsiveness. Solvent effects increase α_{iso} , at
 11 the B3LYP level, it rises from 47.51×10^{-24} esu in vacuum to 59.45×10^{-24} esu with PCM and
 12 62.19×10^{-24} esu with SMD, corresponding to increases of 25.1% and 30.9%, respectively.
 13 Similar trends are observed for all functionals, with the larger SMD values highlighting the
 14 contribution of solvent-induced polarisation and nonelectrostatic solute–solvent interactions.
 15 α_{iso} also increases slightly with increasing field frequency. At the B3LYP level, α_{iso} increases
 16 from 47.73×10^{-24} esu at 1906 nm to 48.25×10^{-24} esu at 1064 nm. Comparable trends are
 17 found for PCM and SMD calculations, although the overall variation remains below 2%,
 18 confirming that the studied wavelengths lie within the nonresonant regime.

19 The anisotropic polarisability follows trends similar to those observed for α_{iso} , although with
 20 smaller absolute values (**Figure 10D**). In the static gas phase, $\Delta\alpha$ ranges from 10.21 to

1 10.86×10^{-24} esu, depending on the functional. Compared with *p*-nitroaniline (7.5×10^{-24}
 2 esu), OPPh₃py exhibits anisotropic polarisabilities approximately 1.4 times larger, indicating a
 3 stronger directional electronic response. In contrast to α_{iso} , solvent effects slightly reduce $\Delta\alpha$. At
 4 the B3LYP level, $\Delta\alpha$ decreases from 10.86×10^{-24} esu in vacuum to 10.42×10^{-24} esu (PCM)
 5 and 10.55×10^{-24} esu (SMD). This reduction suggests that solvent polarisation promotes a more
 6 homogeneous electronic density distribution, thereby diminishing the anisotropy of the
 7 molecular response. A modest frequency dependence is also observed. For example, the
 8 SMD/B3LYP value increases from 11.42×10^{-24} esu at 1906 nm to 11.75×10^{-24} esu at 1064
 9 nm, corresponding to an enhancement of about 2.9%. Similar behaviour is found for all
 10 investigated methods, reflecting the increasing contribution of electronic polarisation as the
 11 applied field approaches resonance conditions. B3LYP consistently predicts the largest α_{iso} and
 12 $\Delta\alpha$ values, whereas CAM-B3LYP and M06-2X yield lower estimates. The range-separated
 13 ω B97XD functional provides intermediate results, while the differences between M06-2X and
 14 *M06-2X are negligible, indicating a limited basis-set effect on the calculated linear
 15 polarisabilities.

16 3.4.2 The second-order NLO properties

17 The second-order NLO response of OPPh₃py was assessed in terms of its first-order
 18 hyperpolarisability, β : [static $\beta(0; 0, 0)$, dynamic $\beta^\lambda(-\omega; \omega, 0)$], and second-harmonic generation
 19 behaviour, β_{SHG} [$\beta^\lambda(-2\omega; \omega, \omega)$], both of which are widely used to evaluate the efficiency of
 20 molecular frequency-doubling processes and charge transfer-induced optical nonlinearity [23].
 21 In push-pull systems, a larger first-order hyperpolarisability indicates enhanced electronic
 22 polarisation under an applied field. The SHG response directly reflects the efficiency of
 23 frequency-doubling and is therefore a key parameter for second-order photonic applications
 24 [13,14,21]. The magnitude of total first-order hyperpolarisability (β_{tot}) was calculated via Eq. 3.17
 25 [85],

$$\beta_{\text{tot}} = \left(\beta_x^2 + \beta_y^2 + \beta_z^2 \right)^{1/2} \quad (3.17)$$

26 where, $\beta_x = \beta_{xxx} + \beta_{xyy} + \beta_{xyz}$, $\beta_y = \beta_{yyx} + \beta_{yyy} + \beta_{yzz}$ and $\beta_z = \beta_{zxx} + \beta_{zyy} + \beta_{zzz}$.

27 **First-order hyperpolarisability (β_{tot}):** The β_{tot} results (Table S7 and Figure 11A) reveal a
 28 pronounced second-order nonlinear optical response for OPPh₃py. In the gas phase, β_{tot} ranges
 29 from 8.71 to 10.30×10^{-30} esu, with CAM-B3LYP predicting the largest values and B3LYP the
 30 smallest. For all functionals, β_{tot} increases as the wavelength decreases from 1906 to 1064 nm,
 31 indicating a progressive enhancement of the nonlinear response as the applied field approaches
 32 the electronic excitation region. Solvation further amplifies the hyperpolarisability, increasing
 33 the static β_{tot} values to 15.38 – 17.73×10^{-30} esu in PCM and 17.12 – 19.89×10^{-30} esu in SMD.
 34 The largest response is obtained at the SMD/CAM-B3LYP level, where β_{tot} reaches
 35 19.89×10^{-30} esu, while the similar values predicted by SMD/M06-2X and SMD/*M06-2X
 36 indicate good consistency across the long-range corrected and meta-hybrid functionals.

37 These trends highlight the strong influence of both frequency dispersion and solvent
 38 stabilisation on the nonlinear optical properties of OPPh₃py. The systematic increase in β_{tot} at
 39 shorter wavelengths reflects the growing contribution of resonance-assisted electronic
 40 polarisation, whereas the larger values obtained in solution demonstrate the ability of the
 41 solvent environment to stabilise polarised charge-separated states. The stronger enhancement
 42 predicted by SMD relative to PCM further suggests that nonelectrostatic solute–solvent
 43 interactions play an important role in promoting electronic delocalisation. Tensor analysis (Table
 44 S7) shows that the dominant contributions arise from the β_y and β_z components, indicating that
 45 the nonlinear response is highly anisotropic and governed by directional intramolecular charge
 46 transfer. The combination of large β and favourable off-resonance optical responses ($\lambda > 1000$
 47 nm) highlights the potential of organic molecules for nonlinear optical and electro-optic
 48 applications [86].

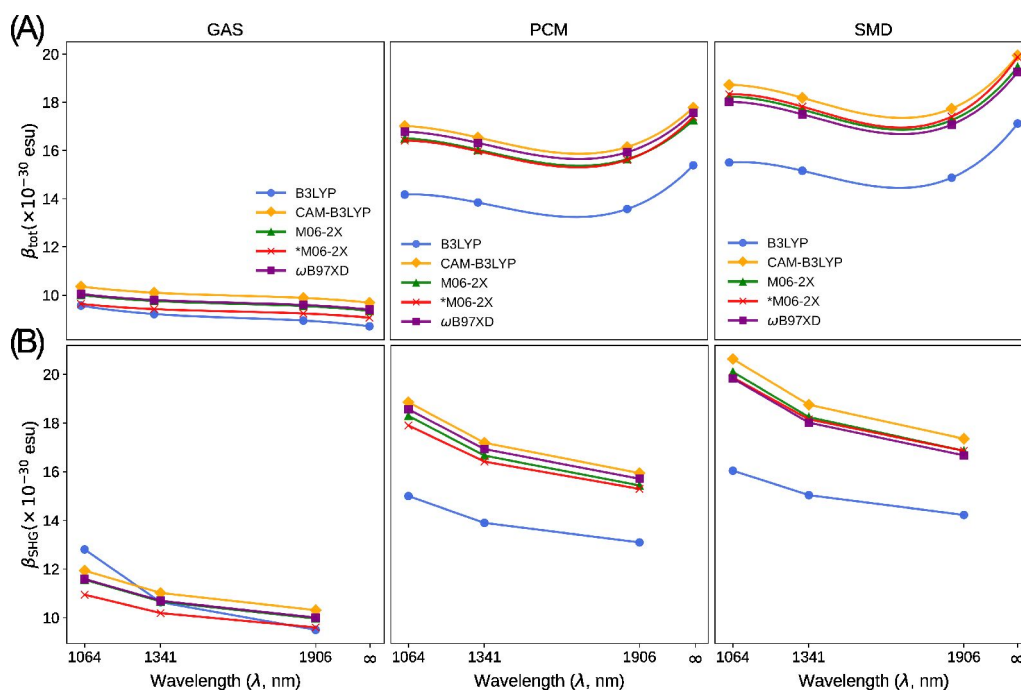


Figure 11. Comparison of (A) static and dynamic first-order hyperpolarisabilities (β_{tot}) and (B) second-harmonic generation hyperpolarisabilities (β_{SHG}) for OPPh₃py computed using the GAS, PCM, and SMD solvation models at various levels of theory.

Second-harmonic generation (β_{SHG}): The β_{SHG} hyperpolarisabilities ($-2\omega; \omega, \omega$) reveal that the SHG response of OPPh₃py is highly sensitive to both the electronic structure method and the surrounding environment (Table S8 and Figure 11B). In the gas phase, β_{SHG} ranges from 9.50 to 12.80×10^{-30} esu, depending on the functional and excitation wavelength. For all investigated methods, the SHG response increases as the wavelength decreases from 1906 to 1064 nm, demonstrating a clear frequency-dispersion effect. While B3LYP predicts the largest gas-phase values, CAM-B3LYP, M06-2X, *M06-2X, and ω B97XD exhibit similar wavelength-dependent trends. Solvation substantially amplifies the SHG activity, increasing β_{SHG} to 13.09 – 18.80×10^{-30} esu in PCM and 14.22 – 20.57×10^{-30} esu in SMD. The strongest response is obtained at the SMD/CAM-B3LYP level at 1064 nm, where β_{SHG} reaches 20.57×10^{-30} esu. The SMD/CAM-B3LYP, SMD/M06-2X, and SMD/ ω B97XD results show close agreement, suggesting a robust description of the SHG response.

The observed trends highlight the combined influence of frequency dispersion and solvent stabilisation on the SHG efficiency of OPPh₃py. The systematic increase in β_{SHG} at shorter wavelengths reflects the growing interaction between the optical field and the molecular charge transfer manifold, leading to enhanced second-order polarisation. The additional amplification observed in PCM and, particularly, in SMD indicates that solvent-induced stabilisation of polarised electronic configurations facilitates charge redistribution during the SHG process. This interpretation is supported by the tensor analysis (Table S8), which shows that the dominant contributions arise from the β_y and β_z components. The predominance of these terms reveals a strongly anisotropic response governed by directional intramolecular charge transfer.

3.4.3 The third-order NLO properties

The third-order nonlinear optical properties of OPPh₃py were investigated through its second-order hyperpolarisability and electric field induced second-harmonic generation [ESHG,

$\gamma_{\text{ESHG}}(-2\omega; \omega, \omega, 0)$ response, which are key parameters for assessing higher-order electronic polarisation under an applied field [16,87]. These properties provide insight into the ability of the molecule to support nonlinear charge redistribution, optical limiting, and other intensity-dependent photonic processes, making them particularly relevant for advanced optoelectronic and all-optical switching applications [80]. In donor- π -acceptor systems, enhanced third-order responses are typically associated with extended conjugation, efficient intramolecular charge transfer, and strong field-induced electron delocalisation. The second-order hyperpolarisability γ was computed utilizing Eq. 3.18 [85],

$$\gamma = \sqrt{\gamma_x^2 + \gamma_y^2 + \gamma_z^2} \quad (3.18)$$

where, $\gamma_i = \frac{1}{15} \sum_{i,j} (\gamma_{ijji} + \gamma_{ijij} + \gamma_{iijj})$ $i, j = \{x, y, z\}$.

Second-order hyperpolarisability (γ): The static $\gamma(0; 0, 0, 0)$ and dynamic $\gamma^\lambda(-\omega; \omega, 0, 0)$ second-order hyperpolarisabilities (Table S9 and Figure 12A), demonstrate that the third-order nonlinear optical response of OPPh₃py is strongly influenced by the electronic structure method, excitation wavelength, basis set, and solvent environment. In the gas phase, γ ranges from 32.26 to 58.62×10^{-36} esu, depending on the computational approach and optical frequency. All functionals predict a systematic increase in γ as the wavelength decreases from 1906 to 1064 nm, indicating enhanced third-order polarisation at higher frequencies. B3LYP consistently yields the largest responses, whereas CAM-B3LYP, M06-2X, *M06-2X, and ω B97XD provide lower but closely comparable values. Solvation further amplifies the third-order response, increasing γ to 55.33 – 95.59×10^{-36} esu in PCM and 56.86 – 98.75×10^{-36} esu in SMD. The largest value is obtained at the SMD/B3LYP level, where γ reaches 98.75×10^{-36} esu, nearly twice the corresponding gas-phase result.

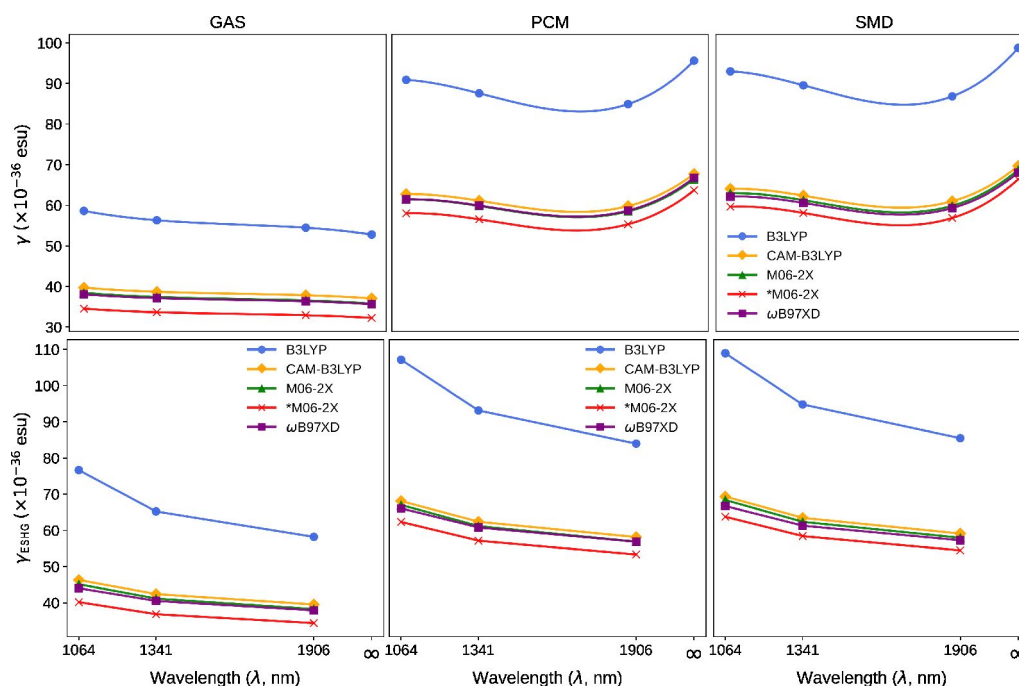


Figure 12. (A) Static and dynamic second-order hyperpolarisabilities (γ) and (B) electric field induced second-harmonic generation responses (γ_{ESHG}) of OPPh₃py computed using the GAS, PCM, and SMD solvation models at various levels of theory.

The magnitude of γ reflects the ability of the electronic density to undergo higher-order polarisation under an external electric field. The systematic increase observed at shorter wavelengths indicates that the electronic cloud becomes progressively more susceptible to optical perturbation as the applied frequency approaches the electronic excitation region. Solvation further enhances this response by stabilizing polarised electronic configurations, thereby facilitating electronic delocalisation and increasing molecular polarisability. The larger values predicted by SMD relative to PCM suggest that nonelectrostatic solute–solvent interactions contribute appreciably to the third-order response. Additional insight is provided by the tensorial analysis, which identifies the diagonal components γ_{xxxx} , γ_{yyyy} , and particularly γ_{zzzz} as the dominant contributors. The predominance of γ_{zzzz} indicates that the third-order response is strongly anisotropic and concentrated along the principal molecular charge transfer direction, consistent with an extended and highly polarisable electronic distribution.

Electric field induced second-harmonic generation (ESHG): The ESHG hyperpolarisability, also known as dc-Kerr effect [88] demonstrate that the third-order nonlinear optical response of OPPh₃py is strongly affected by the electronic structure method, excitation wavelength, basis set, and solvent environment (Table S10). In the gas phase, γ_{ESHG} ranges from 34.37 to 76.66×10^{-36} esu, depending on the computational approach and optical frequency. For all investigated functionals, the ESHG response increases as the wavelength decreases from 1906 to 1064 nm, indicating enhanced third-order polarisation at higher frequencies. B3LYP consistently predicts the largest values, whereas CAM-B3LYP, M06-2X, and ω B97XD yield lower but closely comparable responses. Solvation further amplifies the ESHG activity, increasing γ_{ESHG} to $53.33\text{--}107.15 \times 10^{-36}$ esu in PCM and $54.47\text{--}108.95 \times 10^{-36}$ esu in SMD. The maximum response is obtained at the SMD/B3LYP level at 1064 nm, where γ_{ESHG} reaches 108.95×10^{-36} esu, nearly twice the corresponding gas-phase value. The close agreement between the SMD/CAM-B3LYP and SMD/M06-2X results indicates a consistent description of the ESHG response across the different long-range corrected and meta-hybrid functionals.

Unlike the second-order hyperpolarisability, the ESHG response reflects the coupling between the molecular dipole moment and the field-induced nonlinear polarisation. Consequently, its magnitude is particularly sensitive to environmental effects that stabilise charge-separated electronic configurations. The substantial enhancement observed in solution, especially within the SMD framework, suggests that solvent-induced polarisation promotes more efficient charge redistribution under an applied electric field. Tensorial analysis further reveals that the response is dominated by the diagonal components γ_{xxxx} , γ_{yyyy} , and particularly γ_{zzzz} . The predominance of γ_{zzzz} indicates that the ESHG activity is highly anisotropic and concentrated along the principal molecular charge transfer direction, while its systematic increase in PCM and SMD environments confirms the important role of solvent stabilisation in reinforcing the third-order nonlinear optical response. Organic chromophores exhibiting large off-resonance ESHG responses are attractive candidates for electro-optic materials and devices [89–91].

The overall study shows that OPPh₃py exhibits a pronounced and consistently enhanced optical response arising from its strong donor- π -acceptor architecture and efficient intramolecular charge transfer. The sizeable dipole moment and large isotropic polarisability confirm a highly delocalised and easily polarisable electronic structure, while the moderate anisotropy reflects a well-defined directional charge transfer pathway. This behaviour is further amplified in the nonlinear regime, where both second- and third-order hyperpolarisabilities are characterised by large magnitudes, clear frequency dispersion, and strong solvent dependence. The enhancement observed under SMD conditions highlights the critical role of solute–solvent interactions in stabilizing polarised electronic states and facilitating charge redistribution. The dominance of specific tensor components across β and γ responses confirms that the optical activity is highly anisotropic and aligned along the principal molecular axis. The FMO, NBO, and UV–Vis results provide useful guidelines for the future design of improved NLO

chromophores. Enhancing the intramolecular charge transfer pathway between the donor and acceptor units may further increase the NLO response. Modification or extension of the π -conjugated bridge appears to be the most promising strategy. The calculated electronic structure indicates that electron delocalisation along this pathway plays a dominant role in the observed hyperpolarisability [92]. Increasing the conjugation length, improving planarity, or introducing substituents that restrict rotational freedom may facilitate more efficient charge transfer, reduce the HOMO–LUMO energy gap, and promote electronic polarisation [15,16].

4. Conclusion

In this paper, we studied the geometric-electronic relationships underlying the NLO properties of OPPh₃py using an integrated experimental and computational approach. The optimised molecular structure shows geometrical agreement with the experimental crystallographic parameters. Frontier molecular orbitals, molecular electrostatic potential, and global reactivity analyses demonstrate a well-defined donor- π -acceptor architecture. NBO calculations reveal strong hyperconjugative interactions that promote electronic delocalisation, a key feature of push-pull systems that enhances first- and second-order hyperpolarisabilities. Static and dynamic NLO analyses further demonstrate that OPPh₃py possesses a large dipole moment and high polarisability, thereby increasing the efficiency of intramolecular charge transfer. The static and dynamic polarisabilities and first-order hyperpolarisabilities of OPPh₃py were found to be several-fold greater than those of the standard *p*-nitroaniline molecule calculated at the DFT level. Further analysis of the third-order NLO properties, including the second-order hyperpolarisability and electric field induced second-harmonic generation, confirmed the potential of OPPh₃py for applications in nonlinear optics. Moreover, this study establishes a framework for understanding structure-property relationships in organophosphorus compounds. Future work will involve experimental validation of the predicted NLO properties of OPPh₃py using hyper-Rayleigh scattering, ESHG, and Z-scan measurements, alongside assessments of photostability, thermal robustness, and thin-film fabrication. These studies will provide the foundation for subsequent integration into prototype nonlinear optical devices, such as waveguides and electro-optic modulators. Concurrent computational investigations of related analogues will support molecular optimisation and performance enhancement.

Authors' Contributions: UCS: conceptualisation, data curation, formal analysis, investigation, methodology, project administration, validation, visualisation, writing-original draft, writing-review & editing. LFBV: data curation, investigation, methodology. FCC: methodology. AGR: conceptualisation, investigation, methodology, software. CDL: investigation, writing - original draft, writing-review & editing. ASL: conceptualisation, data curation, formal analysis, investigation, methodology, project administration, software, validation, visualisation, writing-original draft, writing-review & editing, supervision.

Disclaimer: There are no conflicts of interest to declare.

Data Accessibility: The supporting information provides data associated with the proposed reaction mechanism for the synthesis of OPPh₃py (Section S1 and Scheme S1), calculated and experimental geometrical parameters (Tables S1-S2), molecular structures of the reference compounds used for benchmarking (Table S3), local reactivity parameters (Table S4), second-order perturbation theory analysis of Fock matrix (Table S5), and nonlinear optical properties including dipole moments, polarisabilities, hyperpolarisabilities, and related tensor components (Tables S6–S10).

Raw data for single-point energy and NLO calculations are available in <https://doi.org/10.5281/zenodo.18866407>.

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